Gunavardhan Tigulla

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LinkedIn | GitHub | LeetCode | HackerRank

PROFESSIONAL SUMMARY

Aspiring software Developer with strong skills in Python, SQL, and software development, supported by hands-on experience in developing AI-powered applications, REST APIs, and automation solutions. Familiar with Flask, Generative AI, Machine Learning, cloud technologies, and database-driven application development. Strong foundation in object-oriented programming, data structures, and problem solving, with a passion for building scalable software solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, JavaScript
- **Backend & APIs:** Flask, FastAPI, REST APIs, JSON, Automation Scripting
- **AI / ML:** Generative AI, LangChain, LangGraph, RAG, Prompt Engineering, Machine Learning, Deep Learning
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow
- **Databases:** MySQL, PostgreSQL, MongoDB, ChromaDB
- **Tools & Practices:** Docker, Git, SDLC, Agile Methodology
- **Core CS:** OOP, DBMS, Data Structures and Algorithms

PROJECTS

AI-Powered ATS Resume Analyzer *Python, Flask, Generative AI, LangChain, MongoDB, Docker*

- Built a **Flask REST API** that compares resumes against job descriptions using **Generative AI** and **NLP**, returning structured match insights and improvement suggestions.
- Designed **prompt-engineering** workflows with **LangChain** to generate personalized, role-specific recommendations for improving each resume.
- Stored resume data and analysis results in **MongoDB** to support structured retrieval and repeat comparisons.
- Containerized the application with **Docker** for portable, scalable deployment.

Seat Seeker – Intelligent Booking Agent *Python, LangChain, LangGraph, ChromaDB, MySQL*

- Built a **multi-agent** booking assistant with **LangGraph** that automates seat discovery and reservation management.
- Implemented **semantic search** and **Retrieval-Augmented Generation (RAG)** with **ChromaDB** to deliver context-aware recommendations and streamline bookings.
- Used **LangChain** to connect the agents with tools and a **MySQL** database for reliable seat availability and reservation records.

Explainable Deep Learning Intrusion Detection System *Python, TensorFlow, GAN, SHAP*

- Developed a **deep learning** intrusion detection model in **TensorFlow** to classify network traffic for cyber threat analysis.
- Used a **GAN** to generate synthetic samples and enhance the training dataset, improving coverage of attack patterns.
- Applied **SHAP** to explain model predictions and identify which features drove each detection.
- Made detection decisions interpretable, helping analysts understand and trust why traffic was flagged.

EDUCATION

SRM Institute of Science and Technology, Chennai 2022 – 2026
Bachelor of Technology in Computer Science and Engineering CGPA: 7.03 / 10

CERTIFICATIONS

- Python Programming Certification
- Artificial Intelligence and Machine Learning Certification
- SQL and Database Management Certification
- Full Stack Development Certification